

Research article

Appraising differential capabilities of new graduate nurses: Development across the first nine months of employment

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ABSTRACT

Background: Effective transition to practice for new graduate nurses (NGNs) is of national and international concern. Development of NGNs expands well beyond higher education studies and relies heavily on support during their first year of employment. Little is known of the differential development of NGNs, namely the trajectory of developing capability.

Aim: This study differentiates NGN development during their first year of employment through appraisal of universal domains of nursing practice relevant to the international community.

Methods: Using a quantitative evaluation design NGN performance was appraised at 1-, 3-, and 9-months from February 2020 to November 2021, using a 23-item appraisal tool and accompanying behavioural cues organised around four universal domains of practice: professional values and behaviours; comprehensive nursing practice; organisational capabilities; personal growth and development; and a fifth domain specific to Australia, that is, legislative requirements. Workplace performance was appraised by clinical supervisors and numerically rated according to intensity of coaching required to meet requisite standards of practice.

Results: The shift in rating scores of intensity of coaching required, over three time periods across four key universal domains, were statistically significant ($p < .001$). These findings which indicate the intensity of required coaching for maintenance of standards reduced over the time period suggest advancing NGN capability. The domain representing professional values consistently rated the highest. The domain denoting legislative requirements largely flattened after three months.

Conclusion: These findings corroborate the significant development of NGN capability during the first nine months of employment, especially during the initial three months. Furthermore, they provide empirical evidence that NGNs are most adept at demonstrating professional values; a recognised capability developed through employment during pre-registration studies. Discriminant data is of value to inform both targeted development of individual NGNs and when collated, the education needs of cohorts.

1. Introduction

Transition to practice for New Graduate Nurses (NGNs) is of national and international importance (Kenny et al., 2021) as the first year of practice has been shown to affect NGNs intent to remain in the nursing profession (Gardiner and Sheen, 2016). NGNs are defined as registered nurses (RN), who have gained registration with the indicated regulatory authority as an RN for the first time. NGNs are most vulnerable during this transition period (Cao et al., 2021; Lee and De Gagne, 2022). The need to support NGNs upon their initial employment is crucial for their emotional needs and guiding skill acquisition (Doughty et al., 2018;

Brunelli et al., 2022; Reebals et al., 2022; Kaldal et al., 2022). Guided supervision, clinical coaching and [goal-oriented] feedback that assists continuing development of workplace performance directly contributes to safe, ethical, and competent clinicians, (AlMekawi and El Khalil, 2020; Baumann et al., 2019; Harrison et al., 2020a; Rose and Andersson, 2022) and is imperative for a sustainable nursing workforce.

2. Background

The first year of practice is a critical time for NGN learning and development in their progression to become a proficient RN (Harrison

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et al., 2020b). NGNs report lower levels of confidence which can impact delivery of safe patient care (Herron, 2018; Kaldal et al., 2022). Nationally and internationally, many health care organisations provide support and structured programs during the first year of employment to advance NGN competence and capability (Baumann et al., 2019; Lee et al., 2019; Perron et al., 2019; Walton et al., 2018). Universally, transition-to-practice programs are understood to positively impact satisfaction with employment, retention to the profession, and competence to practice (Charette et al., 2022; Kaldal et al., 2022). Anecdotally, this is largely accepted; yet there is minimal objective evidence.

2.1. Transition-to-practice programs

Transition-to-practice programs (TTPs) assist NGNs to navigate learning required to build clinical and communication skills, critical thinking, time management, care delivery, and assimilate into the profession (Harrison et al., 2020b; Wildermuth et al., 2020). At a practical level this is achieved by structured orientation, clinical coaching, focused education days, debriefing, and targeted feedback to support effective assimilation (Henderson et al., 2015). Offerings of TTPs are dependent on the structure and resourcing of education by health services. Within the Australian context TTPs are delivered by individual health services. There is considerable variation in TTPs' content and length; most are six to twelve months and comprise of provision of support, social activities and learning resources (Masso et al., 2019). These programs are not usually affiliated with higher education providers, however sometimes evidence of learning can be credited toward post graduate study. Individual NGN progress is distinct and therefore guidance specific to their experience should be provided (Brunelli et al., 2022). While it is accepted individual needs vary there is still little known in a collective sense about the optimal length or components of TTP (Charette et al., 2022).

2.2. Requisite standards for NGN performance

The scope and focus of nursing activity is guided by health services and the population they serve. Given the breadth of practice it is important nursing is regulated through minimum standard statements. Professional standards guide the conduct and expectations of nursing practice. The International Council of Nurses (ICN, 2013) identifies regulation contributes to public protection and quality patient care. Programs offering sequential guided learning activities as facilitated in TTPs, alongside timely appraisal and feedback make certain NGNs meet minimum standards for practice and advance their capability.

2.3. Differentiating NGN performance

Nursing standard statements are specific to different countries, however internationally, there is considerable consistency regarding the actual practice of nursing (Ossenberg et al., 2016). Given this consistency it is reasonable to organise standards of practice into domains. Domains comprise a collection of statements that reflect core areas of practice. Four universal areas of practice important for determining student readiness for practice and commensurate with employability have been recognised as: *professional values* (refers to behaviours consistent with respect, integrity, privacy), *organisational capabilities*, *leading and managing in clinical contexts* (refers to ability to work in a team and with broader professional groups), *nursing practice and decision-making* (refers to comprehensive nursing practice including, assessment, planning, delivery and evaluation of care), and *personal growth through scholarship and use of evidence* (refers to on-going learning and education) (Ossenberg et al., 2016; Wise et al., 2022). These areas of practice that are akin to domains have remained relatively consistent across international borders. Primary domains useful for guiding NGN development are clearly illustrated when requisite standards of nursing practice, from geographically separate regions of the world, are

thematically organised (Table 1).

2.4. Appraisal of NGN performance

There is limited rigorous knowledge regarding differential elements of the transitioning NGN progress during initial employment despite continued investment in TTPs. Arguably, this could be attributed to limited reliable and discriminatory appraisal tools of NGN workplace performance. Appraisal on performance in the dynamic practice setting is an essential element of workplace learning (Ossenberg et al., 2020a). Self-reporting tools have been historically utilised by NGNs to measure reflection on skill performance. These self-appraisal tools while potentially beneficial in motivating NGNs to review and be accountable for their learning needs, are restricted in supporting comparisons across NGNs given NGNs have variable perceptions of themselves (Charette et al., 2022). Appraisal on workplace performance has further benefits when it is used to commence a two-way conversation and provision of feedback with a clinical supervisor based on the same criteria (Ossenberg et al., 2020b).

Development of NGN capability during their first year of employment has only partially been explored as valid tools have not been routinely used to sequentially explore workplace performance over time.

3. The study

3.1. Aim

The aim of this study was to differentiate NGN development through their first year of employment, by appraising NGN development across five universal nursing domains at one, three and nine months.

3.2. Method

A quantitative evaluation design was used.

3.3. Ethics

An exemption was gained from the health service's Human Research Ethics Committee (reference number EX/202/QMS/84096).

3.4. Setting

An acute tertiary care metropolitan health care facility in the southeast Queensland, Australia provided the setting for the study. The health care facility is a 1037 inpatient bed hospital with adult services offered in specialised surgical, medical, cancer and rehabilitation. Historically, the facility employs 100 to 170 NGNs annually. The number of NGNs employed each year is dependent on nursing workforce staffing requirements.

TTPs are offered to all NGNs in their first year of practice at this facility. Program offerings are supported by a central education unit in liaison with clinical units; these include guided coaching at the bedside, focused education days, individual appraisal, and feedback sessions. The NGNs are supported and appraised by a range of educationally skilled nurses; we refer to these educationally skilled nurses, in this paper, as 'clinical supervisors'. These clinical supervisors who support and appraise NGNs are employed as nurse educators (NE), a nurse who is accountable at an advanced practice level for the design, implementation, assessment, evaluation of nursing education programs, managing educational resources and providing nursing expertise to support excellence in clinical practice (Queensland Health., 2023), and clinical facilitators (CF), a nurse who has advanced clinical skills, is engaged specifically to provide support, guidance, appraisals, and feedback to nursing staff at the point of patient care. Separate from providing direct clinical care (Queensland health., 2023). Nurse educators and clinical

Table 1
Universal domains mapped across standards from four international jurisdictions.

Universal domains of practice	America Nursing scope and standards of practice ^a	Australia Standards for practice ^b	Japan Clinical nursing competence ^c	United Kingdom Standards of proficiency ^d
Professional values and behaviours	Ethics Culturally congruent practice Communication Professional Practice Evaluation	Thinks critically and analyses nursing practice Engages in therapeutic and professional relationships	Ability to collaborate with other health professionals Ability to build interpersonal relationships	Being an accountable professional
Comprehensive nursing practice	Assessment Diagnosis Outcomes identification Planning Implementation Evaluation	Develops a plan for nursing practice Evaluates outcomes to inform nursing practice	Ability to provide nursing care Ability to ensure and deliver high quality care	Promoting health and preventing ill health Assessing needs and planning care Providing and evaluating care Co-ordinating care
Organisational capabilities	Collaboration Quality of practice Resource Utilisation Environmental health	Provides safe, appropriate, and responsive quality nursing practice	Ability to practice ethically Ability to collaborate with other health professionals	Improving quality and safety
Personal growth and development	Leadership Education Evidence-based practice and research	Maintains the capacity for practice	Ability to apply knowledge	Leading and managing nursing care and working in teams

^a American Nurses Association (2015).

^b Nursing and Midwifery Board of Australia (2016).

^c Matsutani et al., 2010 cited in Fukada (2018).

^d Nursing and Midwifery Council (2018).

facilitators are employed in both the central education unit and specialised clinical units. The NGNs that were appraised in this study were engaged in the TTP as outlined in Table 2. The program on offer recognises the transition-to-practice educational and support requirements for this NGN cohort.

3.5. Sampling and recruitment

Data were collected as part of regular appraisal of NGN performance at one, three, and nine months upon commencement on the unit roster (excluding the requisite orientation and supernumerary period).

3.6. Data collection tool

An existing 23-item appraisal tool, the Australian Nursing Standards Assessment Tool (ANSAT) with accompanying behavioural cues, is currently used by most universities across Australia that offer a Bachelor of Nursing program. It is used to assess nursing student workplace performance against the Nursing and Midwifery Board of Australia (NMBA) registered nurse standards of practice (Nursing and Midwifery Board of Australia [NMBA], 2016). The ANSAT was initially developed in 2013, and was determined to be a valid, reliable, and acceptable tool (Ossenberg et al., 2020a). The ANSAT was reviewed in 2016 to ensure the appraisal tool reflected the changes implemented in the NMBA registered nurse standards for practice (effective June 2016). A non-experimental, two phased approach was used to revise and re-validate the ANSAT 2016 using a modified Delphi approach (Ossenberg et al., 2020a). In phase one minor rewording was required as advised from eight experts, with the content validity ≥ 0.83 (Refer to Supplemental material 2). Phase two field test analysed 7696 completed tools. The revised ANSAT demonstrated high internal consistency (Cronbach's alpha = 0.985) and confirmed to be a valid tool that can be used to measure clinical performance (Ossenberg et al., 2020a).

Given the demonstrated utility and acceptability by both learners and clinical supervisors, it seemed reasonable to adapt for NGN workplace performance appraisal. As all NGNs have met the standards for nursing registration, the descriptors on the rating scale of the ANSAT were adapted to recognise that NGNs have been deemed competent. Accordingly, it was determined to be more useful to ascertain the level of

support and coaching required for NGN transition-to practice and their ongoing progression. Therefore, NGN development was explored in this study using an appraisal scale based on the level of coaching required for the NGN to achieve requirements in statements.

The appraisal scale used was:

- Score of 1 - expected behaviours and practices were not able to be observed and/or assessed;
- Score of 2 - expected behaviours and practices performed at a satisfactory standard with intensive coaching;
- Score of 3 - expected behaviours and practices performed at a satisfactory standard with routine coaching;
- Score of 4 - expected behaviours and practices performed at a proficient standard with routine coaching;
- Score of 5 - expected behaviours and practices performed at an excellent standard with routine coaching.

[Routine coaching is defined as regular supportive clinical direction by a peer or clinical supervisor, in short episodes, for example, 10–15 min per episode when required at the point of patient care. Intensive coaching is defined as constant verbal and or physical direction by a designated clinical supervisor, is required to ensure safe patient care, refer to www.ansat.com.au for further information concerning the levels of coaching].

The adapted tool is known as the *ANSAT for Graduate Registered Nurses* (Supplemental material 1) and has demonstrated validity and reliability (factor loadings and communalities of exploratory factor analysis by ANSAT for Graduate Registered Nurse and ANSAT demonstrated high internal consistency Cronbach's alpha = 0.985, refer to supplemental material 2). It is also supported with a behavioural cues document (Supplemental material 1). The behavioural cues use 'everyday language' to describe practices and actions to indicate if attributes of the domain of the nursing standards are evident. The behavioural cues document also adds to the reliability of the tool as it provides consistency regarding the criteria when judgements are on performance by the clinical supervisor (Ossenberg et al., 2016).

Table 2
Transition to practice program outline.

Activity	Description	Rationale for inclusion in Transition to practice program
Orientation	Three days of general hospital orientation. Program includes professional nursing values and practice, National Safety and Quality Health Standards, ethical practice, and mandatory training. Integrated electronic medical record training. Allocation of a preceptor (a preceptor is a nurse that works clinically and aside NGN to advise, mentor and guide but has their own clinical load). Four days supernumerary time working with allocated preceptor, includes specific clinical unit orientation. (Supernumerary time is not counted as a part of staffing number, workload sharing with a preceptor). Specific clinical unit learning activities and clinical assessments.	Awareness of professional values, behaviours, and legislative requirements. (Wise et al., 2022) Workplace culture expectations. (Kaldal et al., 2022) Provides technical support. (Kaldal et al., 2022)
Monthly debriefing sessions (1 h)	The debriefing sessions are designed to stimulate reflections on transition to practice experiences and communication of clinical events. Providing opportunities for peer support and sharing of experiences.	Emotional support, foster critical thinking. (Kaldal et al., 2022) Building resilience. (Brunelli et al., 2022) Personal growth and development. (Wise et al., 2022) Communication skills development and technical support. (Kaldal et al., 2022)
Focused Education days (four eight-hour days)	Themes of education days include communication for patient safety (e.g., error and harm in health care, communication skills in health care; and escalation process and skills). Management of patient deterioration in the clinical environment (e.g., management of the patient with acute pain and dementia and delirium; clinical assessment skills and immersive simulation of a deteriorating patient). Clinical skills practice. Quality and safety in health care (e.g., application of the quality improvement process to apply evidence-based guidelines to achieve positive outcomes for patients; commencement of a quality improvement initiative). Leadership in health care (e.g., leadership styles in nursing; strengths and capabilities in leadership; NGNs presentation of their quality improvement initiative). Debrief and reflection.	Building comprehensive nursing practice. (Wise et al., 2022) Building organisational capabilities. (Wise et al., 2022) Leadership skills development. (Kaldal et al., 2022)

Table 2 (continued)

Activity	Description	Rationale for inclusion in Transition to practice program
Clinical coaching	Support provided by the clinical supervisors. Provides point of patient care educational interventions including feedback to advance clinical skill and practice development (Faithfull-Byrne et al., 2017).	Support time management skills. (Brunelli et al., 2022) Technical support, advocacy, emergent assistance. (Kaldal et al., 2022)
Formal workplace performance appraisal and feedback	Feedback on clinical practice at one-and three-months to focus on specific needs of NGN learning and development of targeted learning plans. Career development session at nine-month appraisal.	Feedback is an essential element of learning to identify strengths and weakness for professional development. (Ossenberg et al., 2020b)

3.7. Data collection

Data were collected using the ANSAT for Graduate Registered Nurses. NGN performance was recorded at one, three and nine months from February 2020 to November 2021. The appraisal times were selected as these time-points have been identified as critical junctures in the NGN journey (Harrison et al., 2020b).

Quantitative data were generated from the ANSAT for Graduate Registered Nurse. The appraisal scores for NGNs performance were completed by clinical supervisors, nurses responsible for supervision and feedback of NGNs. As previously explained, clinical supervisors are employed as either nurse educators or clinical facilitators and can be located in a central education unit or locally within the specialised unit.

The nurse educator responsible for the program available to the NGNs maintained a record detailing NGN commencement and sent appraisal reminders to the respective clinical supervisors. All appraisal data were collected on a paper-based version of the tool and entered an excel spread sheet via Microsoft forms.

3.8. Data analysis

The non-identifiable aggregate data from completed the ANSAT for Graduate Registered Nurses were entered into a statical and data analysis software, IBM SPSS® Statistics (version 28). No missing values were expected as an electronic survey was used to record scores, with all items required to be entered. A one-way analysis of variance (ANOVA) was performed to compare the mean total for NGN scores and score of each of the universal domain standards for each time-point. Post-hoc comparison using the Tukey HSD test was employed to determine where differences between appraisal times occurred. Statistical significance was established at $p \leq .05$.

4. Results

From the 192 NGNs that commenced during the data collection period, February 2020 to November 2021, 146 (76 %) NGNs were appraised at one month, 140 (73 %) NGNs were appraised at three months, and 123 (64 %) NGN were appraised at nine months. Of the 192 NGNs, 72 (38 %) had appraisal scores at all three time points, 73 (38 %) had scores at two time points, and 47 (24 %) were only appraised once using the ANSAT for graduate registered nurses. This resulted in a total of 409 completed tools across the three time-points, the number of forms completed for each time-point were well-matched. The appraisal tools were not always completed due to the NGN moving out of the facility, insufficient time by the nurse educator or clinical facilitator, or the paper form not being legible. The majority NGNs appraised were female ($n = 150$, 79 %) which is commensurate with national and global trends

(World Health Organisation, 2020). The central education unit CF completed 130 (32 %) appraisals, the specialised unit CF completed 185 (45 %) appraisals and the specialised unit NE completed 94 (23 %) of the appraisals.

For the purpose of analysing and reporting of ratings, the ANSAT for Graduate Registered Nurses assessment items have been organised according to the universal domains of practice (Table 1). The mean NGN score for each item by appraisal time are outlined in Table 3. Median scores were also calculated to compare results. However, median findings did not show different findings than those expressed as mean.

These results demonstrate improvement in NGN performance in all items over time (Fig. 1). The domain with the highest mean was professional values and behaviour at each time point. Interestingly, the domain means for comprehensive nursing practice, organisational capabilities, and personal growth and development were similar and developed relatively consistently across one, three and nine months. The fifth domain, 'legislative requirements' that was organised separately for the Australian context plateaued more than the other four universal domains after the three-month mark.

The results of the ANOVA and post-hoc comparison detected a statistically significant improvement ($p < .001$) in NGN scores for each domain and the total score at each appraisal time (Table 4). Overall, this equated to scores shifting from NGNs meeting satisfactory standards of practice with intensive to routine coaching to standards of practice being satisfactory or proficient with just routine coaching.

The total mean scores of all universal domains recorded by clinical supervisors in the central education unit (conducted by clinical facilitators) ($M = 68.76$, $SD = 15.89$) were lower than the collective rating by clinical supervisors located in the specialised clinical units; that is, clinical facilitator ($M = 79.93$, $SD = 18.80$) and nurse educator ($M = 84.09$, $SD = 16.97$) (Table 5). A pairwise comparison between central education unit clinical supervisors, with clinical supervisors in specialised units, namely, clinical facilitators and nurse educators revealed a statistically significant difference ($p < .001$) when appraising graduate performance. No statistically significant difference of appraisal by clinical supervisors in specialised units regardless of whether they were employed as a clinical facilitator or a nurse educator.

5. Discussion

This study sequentially appraised NGNs performance during their first year of employment through a workplace performance appraisal based on universal domains of nursing practice. As transition is recognised as challenging and stressful (Cao et al., 2021; Lee and De Gagne, 2022) understanding performance and progress is important. Findings from this study provide insights into progressive NGN development supported through facility-based programs.

5.1. NGN development

The consistent improvement of NGN performance from one, three to nine months, across the four universal domains of practice—professional values and behaviours, organisational capabilities, comprehensive nursing practice and personal growth and development—suggest value in a program that comprises focused education days, point of care coaching, timely appraisal and feedback resourced by clinical supervisors available further to workload allocations. Of interest is that professional values and behaviours rated higher on initial employment and remained consistently higher. This is in line with systematic reviews that identified students who were employed in health care and other industries during their studies acquired professional capabilities (Wise et al., 2022; Phillips et al., 2016). As employment during pre-registration studies, it is not a surprise that professional capabilities are rated higher. The plateau of NGN performance for the domain legislative requirements was of interest as it may indicate that after three months there is consolidation, and the essential legislative requirements

Table 3

Universal domains of nursing practice categorised with ANSAT items for graduate registered nurse mean item score by appraisal time for each standard.

Universal domains of nursing practice	ANSAT item ^a	1 month (n = 146, 36 %)	3 months (n = 140, 34 %)	9 months (n = 123, 30 %)
		M^b (SD)	M (SD)	M (SD)
Professional Values and behaviours	1.2 Uses an ethical framework to guide decision making and practice	2.95 (0.74)	3.33 (0.75)	4.11 (0.63)
	1.3 Demonstrates respect for individual and cultural (including Aboriginal and Torres Strait Islander) preference and differences	3.18 (0.67)	3.69 (0.79)	4.54 (0.66)
	2.1 Communicates effectively to maintain personal and professional boundaries	3.23 (0.81)	3.52 (0.79)	4.24 (0.67)
	2.4 Demonstrates respect for a person's rights and wishes and advocates on their behalf	3.05 (0.75)	3.56 (0.82)	4.44 (0.70)
	3.4 Recognises and responds appropriately when own or other's capability for practice is impaired	2.84 (0.83)	3.34 (0.77)	3.84 (0.81)
	4.1 Completes comprehensive and systematic assessments using appropriate and available sources	2.64 (0.78)	3.09 (0.78)	3.88 (0.67)
Comprehensive nursing practice	4.2 Accurately analyses and interprets assessment data to inform practices	2.63 (0.72)	3.14 (0.76)	3.84 (0.66)
	5.1 Collaboratively constructs a plan informed by the patient/client assessment	2.77 (0.80)	3.29 (0.70)	3.85 (0.83)
	6.1 Delivers safe and effective care within their scope of practice to meet outcomes	3.03 (0.67)	3.52 (0.76)	4.03 (0.72)
	6.2 Provides effective supervision and delegates care safely within their role and scope of practice	2.73 (0.78)	3.13 (0.73)	3.94 (0.68)
	6.3 Recognise and responds to practice that may be below expected organisational, legal, or regulatory standards	2.66 (0.76)	3.19 (0.72)	3.80 (0.86)
	7.1 Monitors progress toward expected	2.82 (0.73)	3.31 (0.67)	3.91 (0.64)

(continued on next page)

Table 3 (continued)

Universal domains of nursing practice	ANSAT item ^a	1	3	9
		month (n = 146, 36)	months (n = 140, 34)	months (n = 123, 30)
		M ^b (SD)	M(SD)	M (SD)
Organisational capabilities	goals and health outcomes			
	2.2 Collaborates with the health care team and others to share knowledge that promotes person-centred care	2.93 (0.81)	3.29 (0.79)	4.11 (0.68)
	2.3 Participates as an active member of the healthcare team to achieve optimum health outcomes	2.89 (0.75)	3.36 (0.78)	4.07 (0.73)
	5.2 Plans care in partnership with individuals/ significant others/ health care team to achieve expected outcomes	2.72 (0.73)	3.23 (0.74)	3.88 (0.77)
Personal growth and development	7.2 Modifies plan according to evaluation of goals and outcomes in consultation with the health care team and others	2.72 (0.80)	3.18 (0.71)	3.76 (0.82)
	3.1 Demonstrates commitment to life-long learning of self and others	2.81 (0.89)	3.16 (0.85)	3.98 (0.61)
	3.2 Reflects on practice and responds to feedback for continuing professional development	2.92 (0.80)	3.42 (0.76)	4.00 (0.79)
	3.3 Demonstrates skills in health education to enable people to make decisions and take action about their health	2.67 (0.68)	3.18 (0.76)	3.79 (0.70)
	1.4 Sources and critically evaluates relevant literature and research evidence to deliver quality practice	2.59 (0.71)	3.15 (0.80)	3.93 (0.77)
	Legislative requirements			
Legislative requirements	1.1 Complies and practices according to relevant legislation and local policy	3.18 (0.85)	3.51 (0.72)	4.13 (0.64)
	1.5 Maintains the use of clear and accurate documentation	2.97 (0.78)	3.41 (0.76)	4.12 (0.67)
	3.5 Demonstrates accountability for decisions and actions appropriate to their role	2.87 (0.80)	3.41 (0.79)	4.01 (0.73)

^a Each item rated on 5-point Likert scale:

1 – expected behaviours and practices were not able to be assessed.

2 – expected behaviours and practices performed at a satisfactory standard with intensive coaching.

3 – expected behaviours and practices performed at a satisfactory standard with routine coaching.

4 – expected behaviours and practices performed at a proficient standard with routine coaching.

5 – expected behaviours and practices performed at an excellent standard with routine coaching, presented as a mean/average.

^b Median scores were also calculated to compare results. However, median scores did not show any new findings than those expressed as the mean.

are clearly understood.

5.2. Individual benefits

Individualised feedback is given to each NGN at the time of appraisal focusing on their progression and opportunities for improvement. Feedback is an essential element of learning when constructive; It assists individuals to identify strengths and weakness (Gardiner and Sheen, 2016; Ossenbregt et al., 2020b). The approach to appraisal (i.e., the use of an appraisal tool with specific items that reflect professional nursing standards, that are clarified through behavioural supportive cues), assists clinical supervisors to provide NGNs with specific, timely, respectful, and constructive feedback. Individual benefits include the value of personalised feedback and recognition of specific learning needs during the transition year. Key areas can be identified by clinical supervisors that inform the development of targeted learning plans rather than discussing generic concepts. Shifting the language using behavioural cues can contribute to a positive workplace environment that supports NGNs to develop their practice. This feedback should enhance the quality of care they provide, promote retention in the healthcare workforce and improve health care practice that directly impacts on patient outcomes (Murray et al., 2018).

5.3. Organisational benefits

Collation of individual scores provides evidence that performance improvement was statistically significant at one, three, and nine months. Organisational benefits include collective evidence of the NGN cohorts' improved capability during participation in the learning opportunities provided by the facility. Evidence of the effectiveness of a TTP program is important as hospitals and health services work within fiscal constraints and need to account for funding allocated to support training and development. Fostering NGN capability through coaching provided by clinical supervisors and opportunities to attend focused education days can be a costly exercise. Additionally, it is anticipated that the benefits of this investment of time and money in TTP will increase 'job' satisfaction and decrease the turnover of NGNs in the workforce (Weller-Newton et al., 2022).

Opportunities to collect data from other hospital and health services, nationally and internationally could facilitate benchmarking. Benchmarking can ascertain consistency or variability of NGN workplace capacity during transition to practice depending on support required for learning (Takashima et al., 2019). Domains of sustained poor performance in appraisal items could be shared with higher education providers.

Furthermore, there is potential to explore associations between capability and TTP offerings. The TTP on offer at the acute care metropolitan facility in this study includes dedicated focused education days to engage in targeted learning relevant to practice areas that the hospital deems important to safe patient care. The results from this study suggest that the design and content of the TTP at the study site in conjunction with at the point of care learning and feedback meet the development needs for NGN transition to become proficient in their practice. TTPs are expected to be relevant to the health care requirements and facilitate learning opportunities that are tailored to the

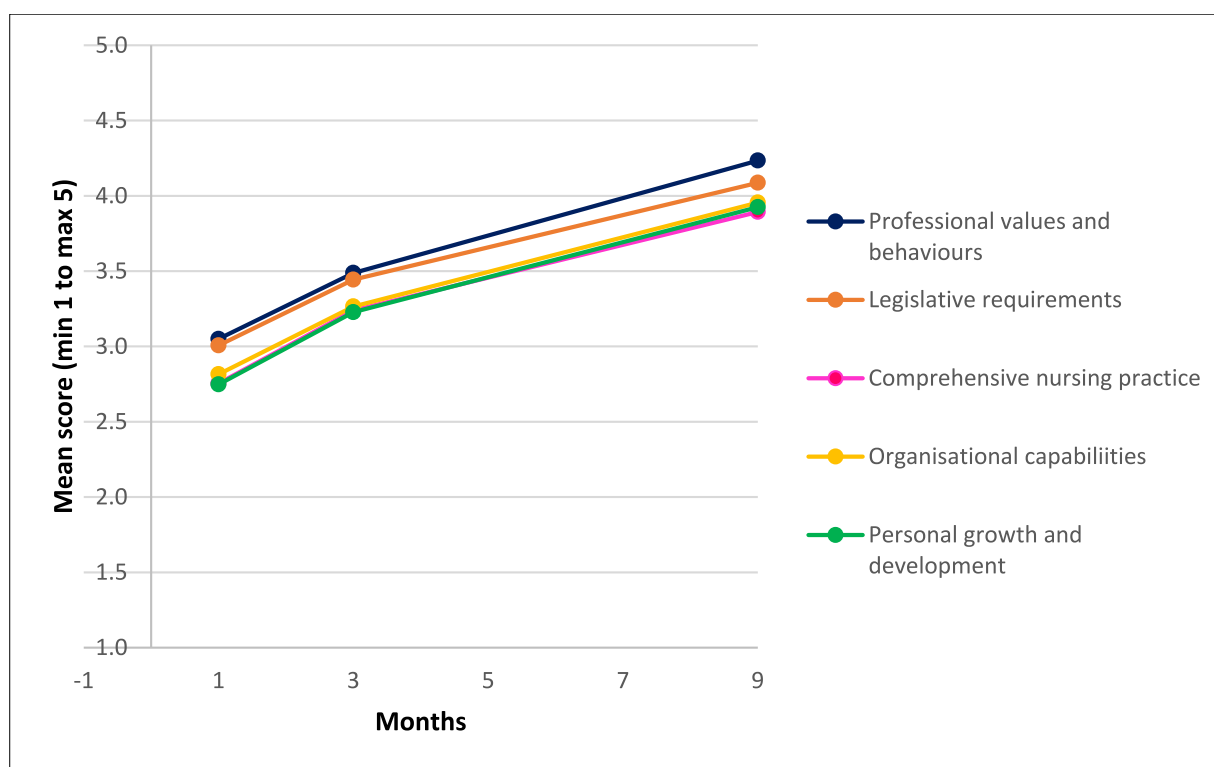


Fig. 1. New graduate nurse mean score for each universal domain of nursing practice over time.

Table 4

Comparison of mean score of each universal domains of nursing practice as categorised with the Australian Nursing Standard Assessment Tool (ANSAT) items by assessment time.

Universal Domains of nursing practice ^a	Mean score ^b						Mean score difference comparison between assessment time periods		
	1 mths (n = 146) M (SD)	3 mths (n = 140) M (SD)	9 mths (n = 123) M (SD)	F	df	P ^c	1 mth vs. 3mth	1 mth vs. 9 mth	3 mth vs. 9mth
Professional values and behaviours	15.23 (3.27)	17.42 (3.33)	21.16 (2.84)	117.42	2	<0.001	-2.18***	-5.92***	-3.73***
Comprehensive nursing practice	19.26 (4.52)	22.66 (4.30)	27.25 (4.21)	112.53	2	<0.001	-3.39***	-7.98***	-4.58***
Organisational capabilities	11.26 (2.76)	13.05 (2.63)	15.81 (2.60)	98.11	2	<0.001	-1.79***	-4.55***	-2.75***
Personal growth and development	10.98 (2.62)	12.91 (2.71)	15.71 (2.44)	110.18	2	<0.001	-1.92***	-4.72***	-2.79***
Legislative requirements	9.02 (2.13)	10.33 (2.01)	12.26 (1.82)	87.58	2	<0.001	-1.31***	-3.23***	-1.92***
Total score	65.75 (15.30)	76.37 (14.98)	92.19 (13.91)	525.82	2	<0.001	-10.59***	-26.40***	-15.77***

*Statistical significance $p < .05$; **statistical significance $p < .01$; ***Statistical significance $p < .001$.

^a See Table 2 for universal domains of nursing practice as categorised with the ANSAT items

Professional values and behaviours (score range: 5–25).

Comprehensive nursing practice (score range: 7–35).

Organisational capabilities (score range: 4–20).

Personal growth and development (score range: 4–20).

Legislative requirements (score range: 3–15).

Total score range: 23–115.

^b Mean scores of ANSAT items as categorised with universal domains of practice.

^c p value calculated using one-way ANOVA with post hoc comparison using Tukey.

identified knowledge deficits. Simply put, TTPs are required to be designed, based on empirical evidence to strengthen NGNs clinical knowledge and skills. Further investigation of the sustainable benefits of TTPs is warranted.

The results indicated significance difference in the appraisals of clinical supervisors based on whether they were located in the central education unit or locally, in the specialised units. These results indicate judgements are possibly based on the perceived purpose of the appraisal, that is congruent results from staff within the same units focused on

appraising need for assistance contrasted with when performance was appraised by an appraiser from a central education unit designed to identify learning opportunities, and possibly, rated lower (Ossenberget al., 2020a, 2020b). This preliminary investigation concerning the difference in appraisals by clinical supervisors warrants further investigation at each time point.

Table 5

Comparison of mean score of each universal domains as categorised with the Australian Nursing Standards Assessment Tool (ANSAT) items for new graduate nurses by assessor role.

Universal domains of nursing practice ^a	Mean score ^b						Mean score difference comparison between groups		
	Central education unit CF (n = 130) M (SD)	Specialised unit CF (n = 185) M (SD)	Specialised unit NE (n = 94) M (SD)	F	df	P	Central education unit CF vs. specialised unit CF	Central education unit CF vs. specialised unit NE	Specialised unit CF vs. specialised unit NE
Professional values and behaviours	16.00 (3.42)	18.28 (4.12)	19.21 (3.54)	22.76	2	<0.001	−2.28***	−3.21***	−0.93
Comprehensive nursing practice	20.18 (4.65)	23.69 (5.36)	24.78 (5.14)	27.13	2	<0.001	−3.51***	−4.60***	−1.08
Organisational capabilities	11.77 (2.86)	13.71 (3.23)	14.35 (3.02)	23.30	2	<0.001	−1.94***	−2.58***	−0.63
Personal growth and development	11.40 (2.83)	13.64 (3.20)	14.22 (2.87)	30.20	2	<0.001	−2.24***	−2.82***	−0.57
Legislative requirements	9.41 (2.13)	10.61 (2.27)	11.53 (2.40)	24.83	2	<0.001	−1.20***	−2.11***	−0.91
Total score	68.76 (15.89)	79.93 (18.80)	84.09 (16.97)	128.22	2	<0.001	−11.17***	−15.32***	−4.12

*Statistical significance $p < .05$; **statistical significance $p < .01$; ***Statistical significance $p < .001$.

CF - clinical facilitator, NE - nurse educator.

^a See Table 2 for universal domains of nursing practice as categorised with the ANSAT items.

Professional values and behaviours (score range: 5–25).

Comprehensive nursing practice (score range: 7–35).

Organisational capabilities (score range: 4–20).

Personal growth and development (score range: 4–20).

Legislative requirements (score range: 3–15).

Total score range: 23–115.

^b Mean scores of ANSAT items as categorised with universal domains of practice.

^c p value calculated using one-way ANOVA, with post hoc comparison using Tukey.

5.4. Limitations

This study was conducted within one facility, during the COVID-19 pandemic outbreak. The pandemic significantly disrupted delivery of health services and their staff, and work undertaken by nurses responsible for the planning and delivery of learning and teaching. Professional development activities were altered as a consequence of physical distancing policies. However, the TTPs were made a priority during the COVID-19 pandemic outbreak. This required some activities to be altered for example, the monthly debriefing sessions were held 'on-line' rather than face to face.

6. Conclusion

NGNs are the foundation of the future healthcare workforce and future generations. Knowing this, supporting the NGN to successfully transition into the workforce is imperative. As demonstrated in this study, a clear structured process for clinical supervisors to appraise NGNs and provide constructive, timely feedback assists in improving the visibility of NGN growth. Insight into the differential elements of the transitioning NGN progress has individual and organisational benefits. These benefits include tailored feedback, targeted individual learning opportunities, allocation of coaching, essential content for educational workshops and evidence of transition outcomes. Opportunities for benchmarking can determine consistency or variability of NGNs workplace capacity. Utilising a 'fit for purpose', validated appraisal tool provides reliable information about workplace capacity of NGNs; this can be used to inform the structure, purpose, and delivery of TTP and clinical support to best meet NGN needs.

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CRediT authorship contribution statement

MW: conceived the paper; entered the data, analysed, and interpreted the data and prepared the manuscript. CO: analysed the data, critically reviewed the manuscript, and suggested revisions to the

manuscript. KS: gathered, entered the data, critically reviewed the manuscript. AH: critically analysed and interpreted the data, reviewed the manuscript, and suggested revisions to the manuscript. All authors read and approved the final manuscript.

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Declaration of competing interest

None of the authors declare a conflict of interest.

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